

## (i) Parallel Shafts:

If the axis of the two shaft are parallel, then the motion is transmitted by three types of gear:

(a) Spur gear (b) Helical gear (c) Double helical gear.

### (a) Spur Gear:

The Spur gear is used to connect two parallel shaft having straight teeth parallel to the axis of the shaft.

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Case 1: If the two gears having external teeth on the outer surface, the two shafts rotate in opposite direction.

Case 2: If the outer wheel having <sup>Internal</sup> ~~inner~~ teeth and inner wheel having external teeth, then the two shafts rotate in same direction.

Spur gears always have line contact.

### (b) Helical Gear:

It is used to connect two parallel shaft having tooth ~~having~~ inclined to axis of the shaft. Two mating helical gears have same helix angle, but the tooth in opposite hands.

### (c) Double Helical Gear:

A pair of helical gears one having right hand helix and other having left hand helix; double helical gears are also known as herringbone.



## (ii) Intersecting Shaft:

The gears are used to connect two intersecting shafts known as bevel gear. If the teeth are straight radial to the point of intersection of shaft is known as straight bevel.

If the teeth are inclined at an angle, it is known as spiral bevel.

## (iii) Non parallel, Non intersecting Shaft:

The gears are used to connect two non parallel and non-intersecting shafts are known as Skew bevel gear.

## (II) Classification based on the peripheral velocity of gear:

(i) Low velocity gear: Velocity  $< 2 \text{ m/s}$

(ii) Medium velocity gear: velocity blw  $3 \text{ m/s}$  and  $15 \text{ m/s}$

(iii) High velocity gear: Velocity  $> 15 \text{ m/s}$

## (III) Classification based on the position of the teeth on the wheel:

(i) Straight  $\rightarrow$  Spur gear

(ii) Inclined  $\rightarrow$  Helical gear

(iii) Curved  $\rightarrow$  Spiral bevel



## (IV) Classification based on type of gearing.

### (i) External gear:

If the gears of two shafts mesh externally with each other known as external gear. Shafts will rotate in opp direction, Small wheel is known as pinion and large wheel is known as Spur wheel.

### (ii) Internal gear:

Two gears mesh internally with each other, then shafts rotate in same direction, again small wheel is known as pinion and the large wheel is known as annular wheel.

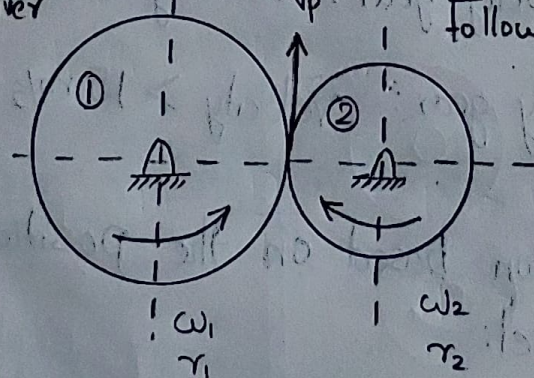
### (iii) Rack and Pinion:

If the gear of a shaft mesh with gear in straight line, the straight line gear is known as rack and the small wheel is known as pinion. Rack and pinion combination Convert rotary motion into linear motion.

Note:

Driver

Follower



$$V = r\omega \quad \text{and} \quad \omega = \frac{2\pi N}{60}$$

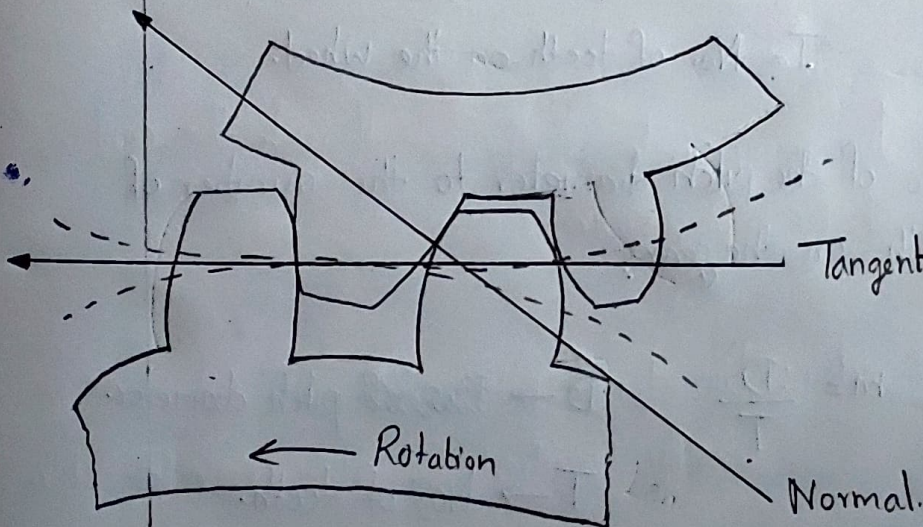


Power transmitted b/w two shafts obtained by two wheels mounted on two shafts. If there is no slip of one surface relative to other, a definite motion of ① can be transmitted to ② and vice versa. Therefore, the tangential velocity also known as pitch velocity at contact surface are same.

Therefore,  $V_p = r_1 \omega_1 = r_2 \omega_2 = 2\pi N_1 r_1 = 2\pi N_2 r_2$

$$\frac{\omega_2}{\omega_1} = \frac{r_1}{r_2} = \frac{N_2}{N_1} = \frac{T_1}{T_2}$$

The relationship shows that, the speed of two disc rolling together without slip are inversely proportional to the radii of disc.



• A pressure angle or Angle of Obliquity:

it is an angle b/w common normal to two gear teeth at the pitch point and common tangent to pitch circle.

Usually denoted by  $\phi$ . Standard pressure angles:  $14\frac{1}{2}^\circ$  &  $20^\circ$